

Notebook: EDA/EDA_breast_cancer_dataset/eda_on_breast_cancer.ipynb

[--- CODE CELL ---]

```
import pandas as pd
```

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```
data = pd.read_csv('data.csv')
```

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```
data.head()
```

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```
# Loaded from CSV instead
```

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```
df = pd.read_csv('../..../datasets/breast_cancer.csv')  
# we set target variables as 0 = malignant, 1 is benign
```

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```
df.head()  
#it is different than the data we imported from kaggle
```

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```
df.shape  
# shows how many rows and columns we have
```

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```
df.columns  
#describes the features of the data
```

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```
df.info()
```

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```
df.isnull().sum()
```

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```
#the above data shows that there exists no missing value present
```

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```
df.describe()  
#we can see that all the features described have a different value attached to it... the mean standard deviation etc changes a lot
```

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```
df['target'].value_counts()
```

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```
df['target'].value_counts().plot(kind = 'bar')
```

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```
features = ['mean radius', 'mean texture', 'mean_smoothness']
#uni variate analysis that is it takes one feature at a time and explores them
```

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```
df[features].hist()
```

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```
import matplotlib.pyplot as plt
```

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```
plt.figure(figsize=(10,5))
df[features].boxplot()
plt.show()
```

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```
#it shows that there are outliers in the data
```

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```
import seaborn as sns
for feature in features:
    sns.boxplot(x='target', y=feature, data=df)
    plt.title(feature)
    plt.show()
```

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```
sns.scatterplot(x='mean radius', y = 'mean texture', data = df)
plt.show()
```

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```
#correaltion analysis
corr = df.drop('target', axis = 1).corr()
plt.figure(figsize=(12,10))
sns.heatmap(corr, cmap = 'coolwarm')
plt.show()
```

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```
df.
```